

Physics.

Q:2(i)

Theory

① A scientific theory is a well-tested explanation of why and how a natural phenomenon occurs.

② It is supported by a large amount of evidence and can be modified with new evidence.

Example :-

Cell Theory

Atomic Theory.

Law

A law describes what happens in a natural phenomenon, often using a mathematical relationship.

It is based on repeated observations and experiments under specific conditions.

Example :-

Hooke's Law

Newton's Law of universal gravitation

Q:2(ii)

①

$$W = mg$$

given :- $m = 55 \text{ kg}$
 $g = 9.8$

$$W = 55 \times 9.8$$

$$W = 539 \text{ N}$$

② weight on moon..

given:-

$$m = 55 \text{ kg}$$

$$g = 1.62 \text{ N/kg}$$

$$W = 55 \times 1.62$$

$$W = 89.1 \text{ N}$$

© Does the mass change?

No, the mass does not change because mass is the amount of matter in an object and remains constant regardless of the gravitational field. Only the weight changes because the moon has a weaker gravitational field than Earth.

Q.3

① Orbital motion:- motion of an object around another object in a circular or curved path due to a gravitational force.

mathematically,

orbital speed $\therefore$ distance travelled over time

for one complete circular orbit:-

$$v = \frac{2\pi r}{T}$$

(b)

given:-

$$r = 6800 \text{ km} \rightarrow 6,800,000$$

$$T = 5440 \text{ s}$$

$$v = \frac{2\pi r^2}{T}$$

$$\frac{2 \times 3.142 \times 6,800,000}{5440 \text{ s}}$$

$$v \approx 7851 \text{ m/s}$$

(c) converting into km/hr :-

$$7851 \times \frac{3600}{1000}$$

$$= 28,263.6$$

$$= 28,264 \text{ km/hr}$$

(d)

The force is gravitational force :-
It's direction is towards the
centre of the earth.

(e) Centripetal force :-
formula :- $F = \frac{mv^2}{r}$

$$= \frac{750 \times (7851)^2}{6,800,000}$$

$$F \approx 6.80 \times 10^3 \text{ N}$$

MCQs ..

1. ~~A~~ C

reason:- option A, B and D are not correct because the question is about significant figures of 0.02080. since ~~option A, B and D~~ ^{option A, B and D} ~~ignoring~~ ^{ignoring} first two zeros we have 2080. so that is why we chose option C.

2. B

because it is not accurate, both precise and accurate or neither both are precise nor it is accurate. It is previous but not exact that result.

3. A

because spring balance is measuring ~~your~~ force, measuring cylinder is used to measure volume in cylinder and vernier calipers are used to measure small lengths.

4. A

$$= 65 \text{ ml} - 42 \text{ ml} \\ = 23 \text{ cm}^3$$

5. C

initial velocity: 0

Final velocity: 35 m/s

acceleration: $a = 9.10 \text{ m/s}^2$

$$a = \frac{v_f - v_i}{t}$$

$$g = \frac{v_f - v_i}{t}$$

$$t = \frac{v_f - v_i}{g}$$

$$t = \frac{35 - 0}{10}$$

$$= \frac{35}{10} \Rightarrow 3.5 \text{ s}$$

6. A

reason:- It does not move with constant speed or accelerating uniformly or decelerating uniformly. It is simply in rest motion.

7. A

$$v_i = 10 \text{ m/s}$$

$$v_f = 10 \text{ m/s}$$

$$t = 15 \text{ s}$$

$$a = \frac{v_f - v_i}{t}$$

$$= \frac{40 - 10}{15}$$

$$= \frac{40}{15}$$

$$\approx 2.67 \text{ m/s}^2$$

8. B

distance :- speed $\times$ time

$$12 \times 25 = 300 \text{ m}$$

9. D

$$\text{Impulse} = mv_f - mv_i$$

$$m = 0.5 \text{ Kg}$$

$$v_i = 20 \text{ m/s}$$

$$v_f = 0 \text{ m/s}$$

$$I = (0.5)(0) - (0.5)(20)$$

$$I = -10 \text{ N.s}$$

10. A

law of conservation of momentum

$$\text{momentum}_i = \text{momentum}_f$$

$$\text{momentum}_i =$$

$$p = 0.01 \times 200 + 1.99 \times 0$$

$$2 \text{ Kg.m/s}$$

momentum =

$$0.01 + 1.99 = 2.00 \text{ kg}$$

$$2 = 2v_i$$

$$1 = m/s$$

$$v_i = 1 \text{ m/s}$$

11. C

$$F = ma$$

$$m = \frac{F}{a}$$

$$= \frac{18}{2} \Rightarrow 9 \text{ kg}$$

12. B

13. A

clockwise moment = anticlockwise moment.

$$F_1 \times d_1 = F_2 \times d_2$$

$$30 \text{ N} \times 2 \text{ m} = F_2 \times 3 \text{ m}$$

$$60 \text{ Nm} = F_2 \times 3 \text{ m}$$

$$\frac{60 \text{ Nm}}{3 \text{ m}} = F_2$$

$$20 \text{ N} = F_2$$

14. D
15. C

16. B

All are advantages but this one is
~~not~~ it's disadvantage.

17. D

$$P = \rho gh$$

$$\frac{1200 \times 10 \times 15}{1800 \text{ Pa}}$$

18.

$$F = kx$$

$$W = mg$$

$$F = W = mg$$

$$4 \times 10 = 40 \text{ N}$$

20cm

$$\rightarrow 0.02 \text{ m}$$

$$F = kx$$

$$k = \frac{F}{x}$$

$$= \frac{40}{0.02}$$

$$2000$$

$$2000 \text{ N/m}$$

19. A

20. D

Hydraulic press :-

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

$$\frac{20}{0.0015} = \frac{F_2}{0.45}$$

$$20 \times 0.45 = 0.0015 F_2$$

$$9 = 0.0015 F_2$$

$$\frac{9}{0.0015} = F_2$$

$$6000 \text{ N} = F_2$$

21. B

22. C

$$P.E = K.E$$

$$E_k = P.E$$

$$= mgh$$

$$= 4 \times 10 \times 10$$

$$= 400 \text{ J}$$

23. A

force = 90 N

$\theta = 60^\circ$

d = 3 m

formula: $W = Fd \cos \theta$

$$= 90 \times 3 (\cos) (60^\circ)$$

$$= ~~120 (0.5)~~$$

$$= 90 \times 3 (0.5)$$

$$= 135 \text{ J}$$

24. D

25. C

26. B

27. D

28. A

29. C

30. D